

JEE MAIN REVISION MASTER: STATISTICS 2026 1

QUESTIONS 1.67% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 10 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Statistics - Core Application	10 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Statistics - Core Application (High Yield — 10 PYQs)

Core Principle: Key problem pattern in Kcet Statistics.

■■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The mean and standard deviation of 100 items are 50 and 4 , respectively then the sum of all squares of the items is..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The mean deviation about the mean for the data 4,7,8,9,10,12,13,17 is..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Let a,b,c,d and e be the observations with mean m and standard deviation S . The standard deviation of the observations $a+k$, $b+k$..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The mean and standard deviation of 100 items are 50 and 4 , respectively then the sum of all squares of the items is

- (A) 250000
- (B) 251600
- (C) 256100
- (D) 265100

Correct Answer: None

Solution Analysis: For 100 items, the mean is $\bar{x}=50$ and the standard deviation is $\sigma=4$. Using the formula $\sigma^2=\frac{\sum x_{(i)}^2-n\bar{x}^2}{n}$ Substitute the given values: $4^2=\frac{\sum x_{(i)}^2-100 \times 50^2}{100}$ $16=\frac{\sum x_{(i)}^2-2500}{100}$ So, $\frac{\sum x_{(i)}^2-2500}{100}=16$ $\sum x_{(i)}^2=2516 \times 100=251600$ Hence, the sum of all squares of the items is 251600 Correct option: Option B.